

Age- and Arm-Stratified Co-expression Network Rewiring in the Murine Kidney During and After Spaceflight

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Background & Objective

Astronauts show elevated post-flight nephrolithiasis risk, and prior renal studies implicate distal nephron remodeling and transporter dysregulation during spaceflight. Standard differential expression (DE) analysis captures shifts in average transcript abundance, but it does not detect changes in how genes are coordinated with one another.

This study asks whether spaceflight reorganizes **kidney co-expression network architecture** in a manner that depends on **age** and **flight arm** and whether these topological changes reveal biology that remains hidden to conventional DE analysis.

We analyzed RRRM-2 / OSD-771, a full-factorial bulk kidney RNA-seq dataset, using a workflow that integrates:

- nephron-segment deconvolution,
- confounder-aware residualization,
- shared-skeleton sample-specific network inference,
- node2vec embedding with Procrustes alignment,
- permutation-based rewiring significance testing, and
- pathway-level interpretation.

Study Design

Age	Young (16 wk) vs Old (34 wk)
Arm	ISS-T vs LAR
Environment Group	FLT, GC, VIV, BSL
Per cell	5 mice
Total	80 female C57BL/6NTac kidneys

Primary contrasts: ISS-T Young (FLT–GC), ISS-T Old (FLT–GC), LAR Young (FLT–GC), and LAR Old (FLT–GC).

This design separates **acute in-flight remodeling** from **post-return recovery/persistence**, while also testing whether older kidneys respond differently from young kidneys.

Analytical Workflow

Phase 0: Deconvolution

MuSiC against a murine kidney single-cell reference to estimate nephron segment proportions.

Phase 1: Residualization

DESeq2-VST expression with technical and compositional confounders removed while preserving biological structure of interest.

Phase 2: Shared topology

A sparse global edge skeleton was learned after within-cell standardization to reduce Simpson's-paradox edge selection and improve cross-condition comparability.

Phase 3: Sample-specific networks

LIONESS-style inference generated contrast-specific weighted networks on the same shared edge set.

Phase 4: Embedding and rewiring

Node2vec embeddings were aligned by orthogonal Procrustes; per-gene rewiring was quantified as cosine-distance shift between matched contrasts.

Phase 5: Significance and interpretation

Focused permutation testing prioritized highly rewired genes; enrichment analysis identified coherent biological programs.

Recommended Figure: Pipeline Schematic

Genome-Wide Rewiring Is Broad and Orthogonal to DE

Spaceflight produced broad remodeling of kidney co-expression structure across all four contrasts. Mean rewiring scores were similar across strata, indicating a **global transcriptomic reorganization** rather than a single isolated pathway effect.

Contrast	Mean Δ	Max Δ
ISS-T Young	0.284	0.515
ISS-T Old	0.287	0.511
LAR Young	0.271	0.513
LAR Old	0.275	0.539

Critically, rewiring magnitude showed **near-zero correlation with** $|\log_2 FC|$, demonstrating that network topology captures a biological dimension that is largely independent of mean-expression change. In other words, genes can maintain similar abundance while being rewired into different regulatory neighborhoods.

Silent Shifters: Network Remodeling Hidden from DE

We define **silent shifters** as genes with:

- high rewiring (top decile),
- low absolute fold change, and
- no strong DE significance.

These genes are especially important because they suggest **regulatory restructuring without obvious expression-level change**. Across the four contrasts, the analysis identified dozens of silent shifters per condition, showing that a substantial fraction of the spaceflight response would be missed by standard transcriptomic screening alone.

Recommended Figure: Rewiring vs Differential Expression

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Rewiring versus $|\log_2 FC|$ across the four primary contrasts. Near-zero Spearman correlation supports the claim that network rewiring is largely orthogonal to classical differential expression.

Recommended Figure: Silent-Shifter Quadrant Plot

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Age- and Arm-Specific Pathway Remodeling

Pathway enrichment revealed that the biological meaning of rewiring depends strongly on both **age** and **flight arm**.

ISS-T Young: ECM and complement-associated remodeling, consistent with acute structural and extracellular adaptation during flight.

ISS-T Old: EMT, steroid biosynthesis, ribosomal, and proteostasis-related signatures, consistent with a more stress-reactive and age-sensitive in-flight response.

LAR Young: PPAR-centered metabolic recovery programs after return to Earth.

LAR Old: PPAR, NOTCH, complement/coagulation, and translational pathways, suggesting a broader recovery state that still retains strong stress and remodeling components.

Together, these results indicate that **young and old kidneys do not recover through the same transcriptomic topology**.

Recommended Figure: Age-Dependent Rewiring

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Age-dependent rewiring comparison in ISS-T kidneys. Scatter and ranked-gene summaries highlight loci whose flight-associated network shifts diverge between young and old animals.

Recovery vs Persistence After Return

Comparison of ISS-T and LAR contrasts separates rewiring that is **transient and flight-specific** from rewiring that **persists after return**. This helps distinguish acute spaceflight stress responses from longer-lasting network remodeling.

A subset of highly rewired genes appears primarily in ISS-T, suggesting changes that resolve after re-entry. Another subset remains rewired in both arms, indicating more persistent systems-level remodeling.

Recommended Figure: ISS-T vs LAR Recovery Plot

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ISS-T versus LAR rewiring comparison showing genes that remain above the diagonal only during spaceflight versus genes whose rewiring persists after return, thereby separating transient and persistent network effects.